CS342 · VISUAL PROGRAMMING

GUI Program Template

The full midterm problem, step by step

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01 The question format

The full program question in midterms 461 and 462 has the same format: a window picture with an input field, result fields, two buttons Convert and Cancel, and error messages.

Required	In code
A class that is the main window	**public class X extends JFrame**
Components defined as attributes	JTextField, JLabel, JButton above the constructor
The button converts	inner class BConvert implements ActionListener
The second button clears	inner class BCancel with setText("")
Error messages	try, throw, catch and JOptionPane.showMessageDialog
The window size	**setSize(450, 100)**

So the template is fixed; only the field names, the formula and the message texts change from one midterm to another.

02 1. Imports and exceptions

```
import javax.swing.*;          // JFrame, JButton, JTextField, JLabel, JOptionPane, JPanel
import java.awt.*;             // FlowLayout, GridLayout, BorderLayout, Container
import java.awt.event.*;       // ActionListener, ActionEvent
```

Each custom exception is a class extending Exception. If the midterm says they are "already defined", you do not write them.

```
class EmptyFieldException extends Exception { }
class InvalidSecondsException extends Exception { }
```

03 2. The class and its attributes

```
public class TimeConverter extends JFrame {
    private JTextField tSec, tHo, tMin, tS;     // each white box
    private JLabel l1, l2, l3, l4;              // each text
    private JButton bConvert, bCancel;          // each button
```

Read the picture and count: each white box is a JTextField, each word a JLabel, each button a JButton. They sit above the constructor so the inner classes can reach them.

04 3. The constructor

```
    public TimeConverter() {
        setTitle("Time Converter");                    // title from the picture
        setSize(450, 100);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JPanel p = (JPanel) getContentPane();
        p.setLayout(new FlowLayout());
        tSec = new JTextField(10);        // 10 = width in columns
        l1   = new JLabel("seconds  =");
        tHo  = new JTextField(3);
        l2   = new JLabel("h :");
        tMin = new JTextField(3);
        l3   = new JLabel("m :");
        tS   = new JTextField(3);
        l4   = new JLabel("s");
        bConvert = new JButton("Convert");
        bCancel  = new JButton("Cancel");
        p.add(tSec); p.add(l1);           // same order as the picture
        p.add(tHo);  p.add(l2);
        p.add(tMin); p.add(l3);
        p.add(tS);   p.add(l4);
        p.add(bConvert); p.add(bCancel);
        bConvert.addActionListener(new BConvert());    // register the listeners
        bCancel.addActionListener(new BCancel());
    }
```

The order inside the constructor is always: window settings → layout → new for each component → add → addActionListener.

With FlowLayout and width 450, the buttons wrap to the second row on their own. To fix each row explicitly, use GridLayout(2, 1) with one JPanel per row (as in the 462 solution).

05 4. The Convert button

```
    class BConvert implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            try {
                String txt = tSec.getText();                          // 1. read
                if (txt.equals("")) throw new EmptyFieldException(); // 2. empty?
                int s = Integer.parseInt(txt);                        // 3. number?
                if (s < 0 || s > 86400) throw new InvalidSecondsException();  // 4.
range?
                int ho  = s / 3600;                                   // 5. compute
                int min = (s % 3600) / 60;
                int sec = s % 60;
                tHo.setText("" + ho);                                 // 6. show
                tMin.setText("" + min);
                tS.setText("" + sec);
            }
            catch (EmptyFieldException ex) {
                JOptionPane.showMessageDialog(null, "Enter the seconds Value to
convert",
                        "Empty Input", JOptionPane.WARNING_MESSAGE);
            }
            catch (NumberFormatException ex) {
                JOptionPane.showMessageDialog(null, "Enter the seconds as a numerical
Value",
                        "Error: Input Format", JOptionPane.ERROR_MESSAGE);
            }
            catch (InvalidSecondsException ex) {
                JOptionPane.showMessageDialog(null, "Enter the seconds Value in the
range [0..86400].",
                        "Error: Invalid Input Value", JOptionPane.ERROR_MESSAGE);
            }
        }
    }
```

Step	Why this order
Empty first	Without it, parseInt("") throws NumberFormatException and shows the number message instead of the empty one
Then parseInt	It throws NumberFormatException by itself; no throw needed
Then the range	It needs the number to compare it
Compute last	Reached only when everything is valid

"" + ho turns the number into text, because setText takes a String. String.valueOf(ho) does the same.



12,500 seconds = 3 hours, 28 minutes and 20 seconds. Check: `3 × 3600 + 28 × 60 + 20 = 12,500`.

06 5. The Cancel button and main

```
    class BCancel implements ActionListener {
        public void actionPerformed(ActionEvent e) {
            tSec.setText("");          // clear every field
            tHo.setText("");
            tMin.setText("");
            tS.setText("");
        }
    }
    public static void main(String[] args) {
        TimeConverter f = new TimeConverter();
        f.setVisible(true);
    }
}
```

07 Conversions that come up

Conversion	Formula	Example
Seconds → h, min, s	**h = s / 3600 · m = (s % 3600) / 60 · sec = s % 60**	12500 → 3 · 28 · 20
cm → m, dm, cm	**m = cm / 100 · dm = (cm % 100) / 10 · c = cm % 10**	253 → 2 · 5 · 3
Minutes → h, min	**h = n / 60 · m = n % 60**	135 → 2 · 15
Days → weeks, days	**w = d / 7 · d2 = d % 7**	17 → 2 · 3

The rule: / between two ints gives the whole number and drops the fraction; % gives the remainder.

08 Mistakes that lose marks

- Forgetting implements ActionListener on the inner class, or misspelling actionPerformed.
- Forgetting addActionListener, so the button does nothing.
- Forgetting setVisible(true), so the window never shows.
- Declaring the components inside the constructor instead of as attributes, so the inner class cannot see them.
- Mixing throw and throws. Inside the code it is throw new X().
- setText(ho) with a number directly; it needs text: setText("" + ho).
- getText() returns a String; do not compare it with ==, use equals("").